Week-6

Q1: Explain the roles of the **Driver**, **Cluster Manager**, and **Executor** in a Spark application.

Q2: How does Spark's **Lazy Evaluation** strategy improve performance when chain-processing large datasets?

Q3: Write a Spark command to read a CSV file located at `"data/source.csv"`, ensuring the first row is treated as a header and `inferSchema` is enabled.

Q4: What is the difference between **CSV** and **Parquet** in terms of storage (row-based vs. columnar) and why does it matter for performance?

Q5: Given a DataFrame `df`, write a query to select the columns `product_id` and `price` where the `category` is 'Electronics'.

Q6: Write the code to "revise" a DataFrame by renaming the column `old_name` to `new_name` and casting the `price` column from a String to a Double.

Q7: How does Spark use the **Lineage Graph (DAG)** to provide fault tolerance if a worker node fails?

Q8: Write a query to filter a DataFrame `df_orders` for rows where the `status` is 'Completed' AND the `amount` is greater than 1000.

Q9: Explain the concept of **Predicate Pushdown** in Parquet and how it affects the amount of data loaded into memory.

Q10: Write a code snippet to add a new column `final_price` which is the `base_price` multiplied by 1.18 (18% tax).

Q11: What is the difference between **Transformations** and **Actions**? Provide two examples of each.

Q12: Write the Spark command to load a Parquet file from `"path/to/input"`, filter out any rows where `user_id` is null, and save the result as a CSV at `"path/to/output"`.

Q13: In Spark Architecture, what is the difference between **Client Mode** and **Cluster Mode**?

Q14: Write a query to filter a dataset for rows where the `region` is 'North' OR the `priority` is 'High'.

Q15: When exploring a dataset, why is it safer to use `.show(5)` instead of `.collect()` on a multi-terabyte dataset?